
Modern Algebra 1

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Problem 1

Proof. (a)

$$\begin{aligned}\phi(f_1, \dots, f_n) \phi(\ell_1, \dots, \ell_n) &= (a_1^{f_1} \cdots a_n^{f_n}) (a_1^{\ell_1} \cdots a_n^{\ell_n}) \\ &= a_1^{f_1} a_1^{\ell_1} \cdots a_n^{f_n} a_n^{\ell_n} \\ &= a_1^{f_1 + \ell_1} \cdots a_n^{f_n + \ell_n} \\ &= \phi(f_1 + \ell_1, \dots, f_n + \ell_n).\end{aligned}$$

Additionally, notice that

$$(f_1, \dots, f_n) + (\ell_1, \dots, \ell_n) = (f_1 + \ell_1, \dots, f_n + \ell_n).$$

Therefore, ϕ is a homomorphism.

(b) First we show $\phi(F) = G$. Notice that for all $a_i \in G \setminus \{e\}$, there exists $(f_1, \dots, f_n) \in F$ such that $f_j = 0$ for all $j \neq i$, and $f_i = 1$. Notice that

$$\phi(0, \dots, 1, \dots, 0) = a_1^0 \cdots a_i^1 \cdots a_n^0 = e \cdots a_i \cdots e = a_i.$$

Therefore, for all $a_i \in G \setminus \{e\}$, there exists $f \in F$ such that $f \mapsto a_i$. Furthermore, note that for $(0, \dots, 0) = e_F \in F$, and $(0, \dots, 0) \mapsto e_G$ by properties of homomorphisms. Therefore, for all $g \in G$, there exists $f \in F$ such that $f \mapsto g$, and ϕ is surjective. It follows that, $G \subseteq \phi(F)$. Finally, note that $\phi(f) \in G$ for all $f \in F$, so $\phi(F) = G$.

We will now apply the first isomorphism theorem. Note that

$$F / \ker \phi \cong \phi(F) = G.$$

By properties of isomorphisms and quotient groups, we have

$$|G| = |F / \ker \phi| = \frac{|F|}{|\ker \phi|}.$$

It follows that

$$|G| |\ker \phi| = |F|.$$

Since p divides $|G|$, p also divides $|G| |\ker \phi|$, and therefore p divides $|F|$.

(c) Since F is a direct product of the groups $\mathbb{Z}_{k_i}$, we have

$$|F| = k_1 \cdots k_n = \prod_{i=1}^n k_i.$$

(d) Note that $\langle g \rangle$ is cyclic with order m . By the fundamental theorem of cyclic groups, since $p|m$, there exists a subgroup $\langle g^{m/p} \rangle$ with order p . It follows that $|g^{m/p}| = p$.

(e) If p is prime and $p|ab$, then either $p|a$ or $p|b$. This can be generalized easily with induction to show if $p|\prod_{i=1}^n a_i$ for $a_i \in \mathbb{Z}$, then there exists some i such that $p|a_i$. Therefore, since p divides the order of F by part (b), and since

$$|F| = \prod_{i=1}^n k_i$$

by part (c), there exists at least one k_i such that $p|k_i$. Since k_i is the order of a_i , there exists $a_i \in G$ such that $p|a_i$. By part (d), there exists an element of order p in G , namely $a_i^{k_i/p}$. ■

Problem 2

(a) Because I don't feel like writing out a lot of subscripts and definitions, if I write $\mathbf{m} \in \mathbb{Z}^n$, then define $\mathbf{m} = (m_1, \dots, m_n)$. Additionally, $a\mathbf{m} = (am_1, \dots, am_n)$.

Let $\phi : \mathbb{Z}^n \rightarrow \mathbb{Z}_8^n$ be the map $g \mapsto g \pmod{8}$. We will show ϕ is a homomorphism. Let $\mathbf{a}, \mathbf{b} \in \mathbb{Z}^n$. It follows that

$$\begin{aligned} \phi(\mathbf{a} + \mathbf{b}) &= \phi(a_1 + b_1, \dots, a_n + b_n) \\ &= (a_1 + b_1, \dots, a_n + b_n) \pmod{8} \\ &\equiv (a_1, \dots, a_n) + (b_1, \dots, b_n) \pmod{8} \\ &= \phi(\mathbf{a}) + \phi(\mathbf{b}). \end{aligned}$$

Therefore, ϕ is a homomorphism. We claim $\ker \phi = a\mathbb{Z}^n$. First, we will show $a\mathbb{Z}^n \subseteq \ker \phi$. Let $\mathbf{z} \in a\mathbb{Z}^n$. By definition, there exists $\mathbf{v} \in \mathbb{Z}^n$ such that $a\mathbf{v} = \mathbf{z}$. Notice that for all components z_i of $\mathbf{z}$, $z_i = av_i$. Since $a|av_i$, $z_i \equiv av_i \equiv 0 \pmod{a}$, and thus $\mathbf{z} \mapsto \mathbf{0}$. Therefore, $\mathbf{z} \in \ker \phi$.

Now we show the converse. Let $\mathbf{v} \in \ker \phi$. It follows that $a|v_i$ for all i , and thus there exist $q_1, \dots, q_n \in \mathbb{Z}$ such that $\mathbf{v} = (aq_1, \dots, aq_n) = a\mathbf{q}$. Therefore, $\mathbf{v} \in a\mathbb{Z}^n$ by definition. Therefore, $\ker \phi = a\mathbb{Z}^n$.

(b) Let $\psi : G \rightarrow \text{Inn}(G)$ be the map $g \mapsto \phi_g$. We will show ψ is a homomorphism. Since $\psi(ab) = \phi_{ab}$ and $\psi(a)\psi(b) = \phi_a\phi_b$, we must show $\phi_{ab} = \phi_a\phi_b$. Notice that

$$\phi_{ab}(x) = abx(ab)^{-1} = abxb^{-1}a^{-1} = \phi_a(bxb^{-1}) = \phi_a\phi_b(x).$$

Therefore, ψ is a homomorphism. Additionally, note that for any $\phi_a \in \text{Inn}(G)$, by definition $a \in G$, and thus $\psi(a) = \phi_a$. Therefore, ψ is surjective. We claim $\ker \psi = Z(G)$. Note that the identity in $\text{Inn}(G)$ is ϕ_e , and $\phi_e(x) = x$. First, we show $Z(G) \subseteq \ker \psi$. Let $a \in Z(G)$. It follows that $ax = xa$ for all $x \in G$, and thus

$$\phi_a(x) = axa^{-1} = aa^{-1}x = x = \phi_e(x).$$

Therefore, $a \mapsto \phi_e$, and $a \in \ker \psi$. It follows that $Z(G) \subseteq \ker \psi$.

Now we show the converse. Let $a \in \ker \psi$. It follows that $\phi_a = \phi_e$, and thus $axa^{-1} = x$ for all $x \in G$. Notice that

$$axa^{-1} = x \implies axa^{-1}a = xa \implies ax = xa.$$

Since this holds for all $x \in G$, $ax = xa$ for all $x \in G$, and thus $a \in Z(G)$. Therefore, $\ker \psi \subseteq Z(G)$, and thus $\ker \psi = Z(G)$.

Problem 3

Proof. (a) Notice that $HN = \{hn \mid h \in H \wedge n \in N\}$. It follows that

$$HN = \bigcup_{h \in H} hN.$$

Since $H \leq G$, if $h \in H$ then $h \in G$. Since $N \triangleleft G$, it follows that $hN = Nh$ for all $h \in H$, and thus

$$HN = \bigcup_{h \in H} hN = \bigcup_{h \in H} Nh = NH.$$

Therefore, $HN = NH$, and HN is a subgroup of G by a previous homework problem. Let $k \in HN$. Since $HN \leq G$, it follows that $k \in G$. Since N is normal in G , $kN = Nk$ for all $k \in HN$. Therefore, $N \triangleleft HN$.

(b) Let $\phi : H \rightarrow HN/N$, and let $g, h \in H$. Since $N \triangleleft HN$ by part (a), it follows that

$$\phi(g)\phi(h) = gNhN = (gh)N = \phi(gh).$$

Therefore, ϕ is a homomorphism.

(c) Let $gN \in HN/N$. Since $g \in HN$, there exist $h \in H$ and $n \in N$ such that $g = hn$. Since $N = eN$ and $n \in N$, by properties of cosets $nN = N$, and thus $hnN = h(nN) = hN$. Since $h \in H$, $\phi(h) = hN$, and thus ϕ is surjective.

(d) First we show $H \cap N \subseteq \ker \phi$. Let $g \in H \cap N$. It follows that $h \in N$, and $\phi(h) = hN$. Since $h \in N$, by properties of cosets $hN = N$. Since the identity of HN/N is $eN = N$, we have $H \cap N \subseteq \ker \phi$.

Now we prove the converse. Let $h \in \ker \phi$. It follows that $\phi(h) = hN = N$. By properties of cosets, $h \in N$. Since $\ker \phi \subseteq H$, $h \in H$. Therefore, $h \in H \cap N$. Therefore, $\ker \phi = H \cap N$.

(e) Since ϕ is surjective by part (c), $\phi(H) = HN/N$. Since $\ker \phi = H \cap N$ by part (d), and since ϕ is a homomorphism by part (b), by the first isomorphism theorem it follows that

$$H/H \cap N = HN/N.$$

■

Problem 4

(a) Let $|G| = p^n$. It follows from the class equation that

$$p^n = |Z(G)| + \sum_{a_i \notin Z(G)} [G : C(a_i)].$$

Lemma 1. Let $a, b, p \in \mathbb{Z}$ such that $p|a$. Then $p|(a + b)$ if and only if $p|b$.

Proof. Let $a, b, p \in \mathbb{Z}$, and let $p|a$. First, suppose $p|b$. It follows that there exist $q, r \in \mathbb{Z}$ such that $a = pq$ and $b = pr$. It follows that

$$a + b = pq + pr = p(q + r).$$

Since $q + r \in \mathbb{Z}$, $p|(a + b)$.

Now suppose $p|(a+b)$. It follows that there exists $q, r \in \mathbb{Z}$ such that $a = pq$ and $a+b = pr$. It follows that

$$\begin{aligned} a+b &= pr \\ \implies pq+b &= pr \\ \implies b &= pr-pq = p(r-q) \\ \implies p|b. \end{aligned}$$

Therefore, if $p|a$, then $p|b$ if and only if $p|(a+b)$. ■

By Lagrange's theorem, $|cl(a)| = |C(a)|$ divides p^n . It follows that $[G : C(a)]$ divides p^n . Since the only divisors of p^n are powers of p , we know

$$\sum_{a_i \notin Z(G)} [G : C(a_i)]$$

is just a sum of powers of p , which we will write as

$$\sum_i p^{e_i}.$$

Notice that $p|p^m$ for all integers $m \neq 0$. Notice that $[G : C(a_i)] = 1$ implies $C(a_i) = G$, and thus $a_i \in Z(G)$. Therefore, p will divide $[G : C(a_i)]$ for all $a_i \notin Z(G)$, and thus

$$p \mid \sum_i p^{e_i}.$$

Notice that $p|p^n$, and thus

$$p \mid \left(|Z(G)| + \sum_{a_i \notin Z(G)} [G : C(a_i)] \right).$$

By lemma 1, it follows that p divides $|Z(G)|$. Since $e \in Z(G)$ implies $|Z(G)| \geq 1$, and since $p \nmid 1$ for all p , we must have $|Z(G)| > 1$.

(b) By the fundamental theorem of finite abelian groups, it suffices to show G is abelian. Since $|Z(G)| > 1$ and $|Z(G)|$ divides p , we must have either $|Z(G)| = p$ or $|Z(G)| = p^2$. For the first case, suppose for purpose of contradiction that $|Z(G)| = p$. Since $Z(G) \triangleleft G$ and G is finite,

$$|G/Z(G)| = \frac{|G|}{|Z(G)|} = \frac{p^2}{p} = p.$$

By one of the corollaries of Lagrange's theorem, $G/Z(G)$ is cyclic. By problem 5 in homework 7, G is abelian, and thus $Z(G) = G$, and thus $|Z(G)| = p^2$, which is a contradiction. Therefore, $|Z(G)| = p^2$.

Notice that if $|Z(G)| = p^2 = |G|$, since $Z(G) \subseteq G$, it follows that $Z(G) = G$, and thus G is abelian.

Problem 5

Definition 1. Some definitions are necessary because this is going to get messy. Let $\mathcal{L}$ be the set of left cosets of $N(H)$ in G . Let $\mathcal{C}$ be the set of subgroups conjugate to H . For all $K \in \mathcal{C}$, let $\mathcal{G}(K)$ be the set of elements in G such that $gHg^{-1} = K$. Let $\mathcal{D}$ be the set of all sets $\mathcal{G}(K)$. Symbolically, our definitions are

$$\begin{aligned}\mathcal{L} &= \{gN(H) \mid x \in G\} \\ \mathcal{C} &= \{gHg^{-1} \mid g \in G\} \\ \mathcal{G}(K) &= \{g \in G \mid gHg^{-1} = K\} \\ \mathcal{D} &= \{\mathcal{G}(K) \mid K \in \mathcal{C}\}.\end{aligned}$$

This proof is so messy that I think it's worthwhile to give a brief outline of the proof.

1. First, we show $\mathcal{D} = \mathcal{L}$. That is, the set of cosets is precisely the set of sets of elements that send H to a distinct conjugate.
2. Next, we define a bijection $\mathcal{C} \rightarrow \mathcal{D}$ by $K \mapsto \mathcal{G}(K)$. It follows that $|\mathcal{C}| = |\mathcal{D}| = |\mathcal{L}| = [G : N(H)]$.

First, we will show $\mathcal{D} \subseteq \mathcal{L}$. Let $\mathcal{G}(K) \in \mathcal{D}$. It follows that $K \in \mathcal{C}$. Fix some $g \in \mathcal{G}(K)$. We want to show $\mathcal{G}(K) \in \mathcal{L}$, namely $\mathcal{G}(K) = gN(H)$. First, we show $\mathcal{G}(K) \subseteq gN(H)$. Let $x \in \mathcal{G}(K)$. Notice that $xHx^{-1} = K$. Additionally, notice that for any $gn \in gN(H)$, $gnHn^{-1}g^{-1} = gHg^{-1} = K$, since $g \in \mathcal{G}(K)$ and $n \in N(H)$. It follows that

$$\begin{aligned}xHx^{-1} &= gHg^{-1} \\ \implies g^{-1}xHx^{-1}g &= H \\ \implies (g^{-1}x)H(g^{-1}x)^{-1} &= H.\end{aligned}$$

Therefore, $g^{-1}x \in N(H)$. By properties of cosets, $gN(H) = xN(H)$. By properties of cosets, $x \in gN(H)$. Therefore, $\mathcal{G}(K) \subseteq gN(H)$.

Now we show the converse. Let $gn \in gN(H)$. Notice that

$$gnH(gn)^{-1} = gnHn^{-1}g^{-1} = gHg^{-1} = K.$$

Therefore, $gn \in \mathcal{G}(K)$. Therefore, $gN(H) = \mathcal{G}(K)$ for at least one fixed $g \in \mathcal{G}(K)$, and thus $\mathcal{G}(K) \in \mathcal{L}$. Therefore, $\mathcal{D} \subseteq \mathcal{L}$.

Now we show $\mathcal{L} \subseteq \mathcal{D}$. Let $g \in G$ such that $gHg^{-1} = K \in \mathcal{C}$, where K is not necessarily a distinct subgroup from H . Let $gN(H) \in \mathcal{L}$. Notice that $g \in \mathcal{G}(K)$. We wish to show $gN(H) = \mathcal{G}(K)$. Let $gn \in gN(H)$. It follows that

$$gnH(gn)^{-1} = gnHn^{-1}g^{-1} = gHg^{-1} = K.$$

Therefore, $gn \in \mathcal{G}(K)$, and $gN(H) \subseteq \mathcal{G}(K)$. Now let $x \in \mathcal{G}(K)$. It follows that $xHx^{-1} = gHg^{-1}$, and thus

$$g^{-1}xHx^{-1}g = H.$$

Therefore, $g^{-1}x \in N(H)$, and by properties of cosets, $gN(H) = xN(H)$. By properties of cosets, $x \in gN(H)$, and $\mathcal{G}(K) = gN(H)$. Therefore, $\mathcal{L} \subseteq \mathcal{D}$.

Let $\phi : \mathcal{C} \rightarrow \mathcal{D}$ be the map $K \mapsto \mathcal{G}(K)$. First, we show ϕ is well defined. Let $aHa^{-1} = bHb^{-1} = K \in \mathcal{C}$ with $a \neq b$. It follows that $a, b \in \mathcal{G}(K)$, and by properties of cosets, $\mathcal{G}(K) = aN(H) = bN(H)$. Therefore, ϕ is well-defined.

Now we show ϕ is surjective. Let $\mathcal{G}(K) \in \mathcal{D}$. By definition, it follows that $K \in \mathcal{C}$. Finally, we show ϕ is injective. Let $\mathcal{G}(K), \mathcal{G}(L) \in \mathcal{D}$ such that $\mathcal{G}(K) = \mathcal{G}(L)$. By definition, there exists $a, b \in G$ such that $K = aHa^{-1}$ and $L = bHb^{-1}$. By properties of cosets, $a, b \in \mathcal{G}(K), \mathcal{G}(L)$. Therefore, $aHa^{-1} = K$ and $aHa^{-1} = L$, and thus $K = L$.

Therefore ϕ is a bijection, and $|\mathcal{C}| = |\mathcal{D}|$. Since $|\mathcal{D}| = |\mathcal{L}|$, it follows that $|\mathcal{C}| = |\mathcal{L}|$. Since $|\mathcal{L}| = [G : N(H)]$, it follows that $|\mathcal{C}| = [G : N(H)]$.